Data Analysis Project

Data Analyst: Marceline Setiawan {Aspiring Data Analyst (Independent Portfolio Project)}

Client/Sponsor: Superstore Management (Simulated Stakeholder)

Purpose:

The purpose of this project is to analyze Superstore's sales, profitability, discount, and return data from 2023 to 2026 to evaluate business performance and identify opportunities for improvement.

The project aims to understand which regions and product categories contribute most to sales and profit, identify products generating low or negative profit, examine the relationship between discount levels and profitability, and investigate return patterns that may indicate potential customer-experience issues.

The ultimate goal is to translate data into actionable business recommendations that support profitable sales growth, more effective discount practices, and improved customer experience.

Scope / Major Project Activities:

Activity	Description
Define the Business Problem	Establish the main business question, project objectives, and supporting analytical questions.
Data Collection and Understanding	Review the Superstore dataset, including order-level sales records, regional manager information, and return records. Understand the available fields, data structure, and analysis period of 2023–2026.
Data Cleaning and Preparation	Inspect data types, missing values, duplicates, date fields, categorical values, and numerical values. Validate data consistency and relationships between datasets.

Data Transformation	Create analytical fields such as Year, Quarter, Month, Weekday, Shipping Days, Discount Band, Returned Flag, Profit Margin, Sales per Unit, and Profit per Unit.
Exploratory Data Analysis	Analyze sales, profit, and profitability across years, regions, categories, sub-categories, customer segments, and products. Examine discount patterns and returns.
Data Visualization and Dashboard Development	Develop interactive Tableau dashboards to communicate business performance, product profitability, discount analysis, and customer-experience findings.
Insights and Recommendations	Interpret analytical results, identify potential business risks and opportunities, and propose practical actions for management consideration.
Documentation and Presentation	Document the project methodology, findings, limitations, and recommendations. Prepare a stakeholder presentation outlining the results and proposed next steps.

This project does not include:

- Forecasting future sales or profit using predictive models or machine learning.
- Implementing business changes, pricing adjustments, or discount-policy changes within Superstore's operational systems.
- Conducting customer surveys, interviews, or independent investigations into the reasons for product returns.
- Proving that discounts, shipping duration, or any other observed factor directly causes changes in profitability or return rates.
- Evaluating product-specific return reasons where the available return records identify returned orders but do not identify the individual returned products.
- Making definitive business decisions about discontinuing products without further review of costs, demand, and strategic importance.

Deliverables:

Deliverable	Description/ Details
Cleaned and Prepared Dataset	A prepared analytical dataset with validated fields, derived variables, and supporting data relationships.

Data Preparation Documentation	Documentation of data-quality checks, cleaning decisions, calculated fields, data transformations, and relevant assumptions.
Tableau Executive Overview Dashboard	An overview of key performance indicators and sales and profit trends, including regional and category performance and geographical analysis.
Tableau Product Profitability Dashboard	Analysis of product-level sales and profit, identification of high-sales and low- or negative-profit products, and sub-category profitability comparisons.
Tableau Discount Analysis Dashboard	Comparison of profitability across discount bands and investigation of the relationship between discount levels and profit performance.
Tableau Customer Experience Dashboard	Analysis of return-rate patterns and shipping duration by return status, with appropriate consideration of data limitations.
Business Insights and Recommendations	A summary of key findings and practical recommendations related to product profitability, discount practices, regional performance, and potential return-related issues.
Project Documentation and Stakeholder Presentation	Written project documentation and a presentation summarizing the business purpose, analytical approach, findings, recommendations, and proposed management actions.

Schedule Overview / Major Milestones:

Milestone	Expected Completion Date	Description/Details
Project Planning	1 October 2026	Defined the business problem, objectives, and supporting analytical questions.
Data Understanding	3 October 2026	Reviewed the dataset structure, fields, source tables, and analysis period.
Data Cleaning and Preparation	8 October 2026	Conducted data-quality checks and prepared the dataset for analysis.
Data Transformation	12 October 2026	Created calculated fields and

		prepared analytical variables.
Exploratory Data Analysis	17 October 2026	Examined business performance, profitability, discount patterns, and returns.
Tableau Dashboard Development	23 October 2026	Developed four dashboards to present the analytical findings.
Documentation and Recommendations	29 October 2026	Finalized project documentation, verified findings, and refined business recommendations.
Stakeholder Presentation	31 October 2026	Prepared the presentation with key findings, recommended actions, and management decisions requested.

*Estimated date for completion:

Target completion: 31 October 2026, assuming the final review of the documentation, dashboard findings, and stakeholder presentation is completed by that date.